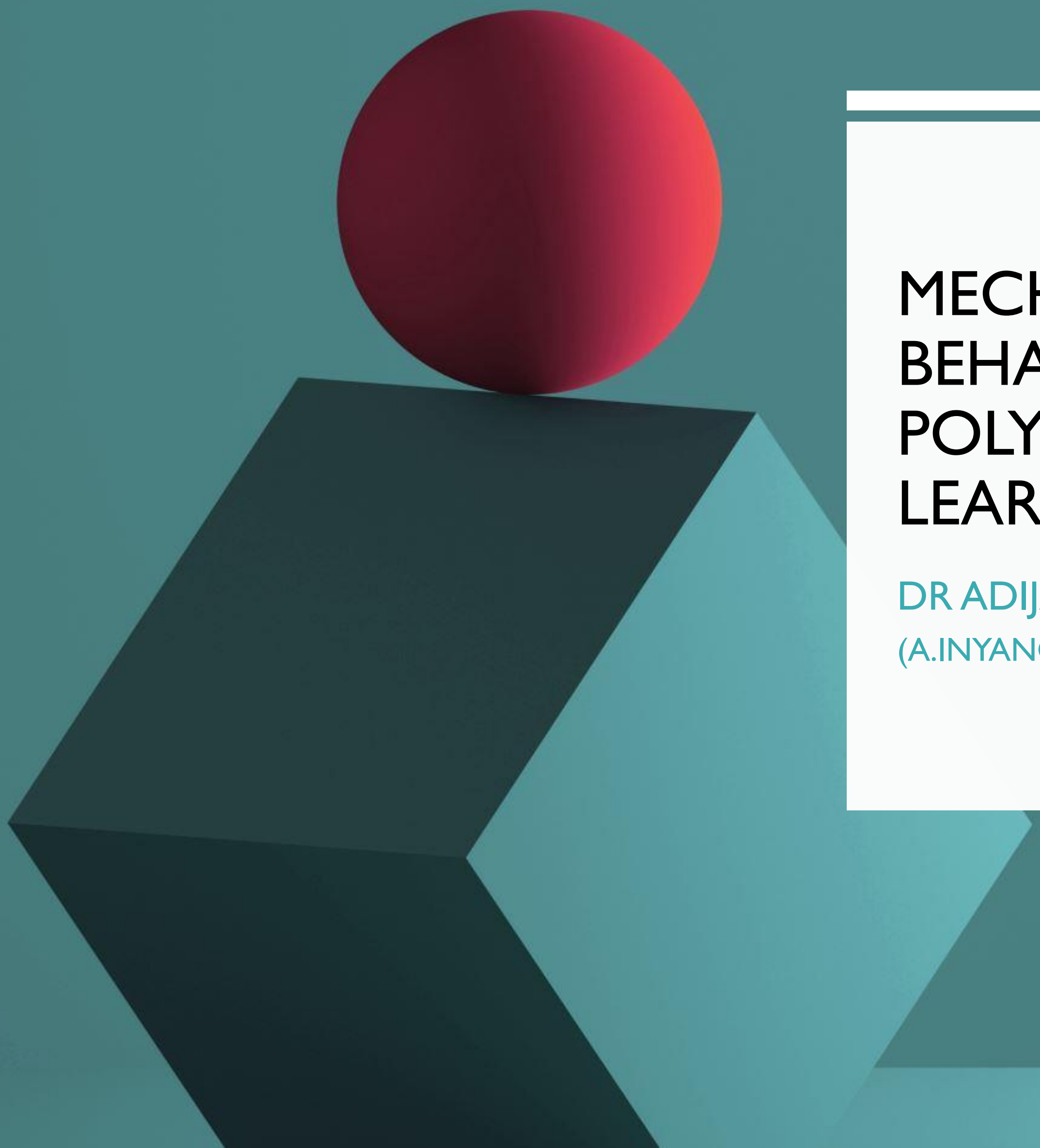

MECHANICAL BEHAVIOR OF POLYMERS – ONLINE LEARNING

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AIM & TASKS

- The aim of this additional learning material is to contribute to your learning of the topics examined in class, related to the mechanical behavior of polymers.
- Complete the questions assigned in every page of this document
- Test your knowledge by taking the mechanical behavior quiz

EXAMPLE 1 – SHORT ANSWER QUESTIONS

- Q1. Practice your writing skills by preparing a short description of the process represented in the schematic representation of the stress-strain behavior of polymeric materials shown in class. Briefly explain every stage of the process for a semi-crystalline polymer. This will help you practice for answering exam questions.

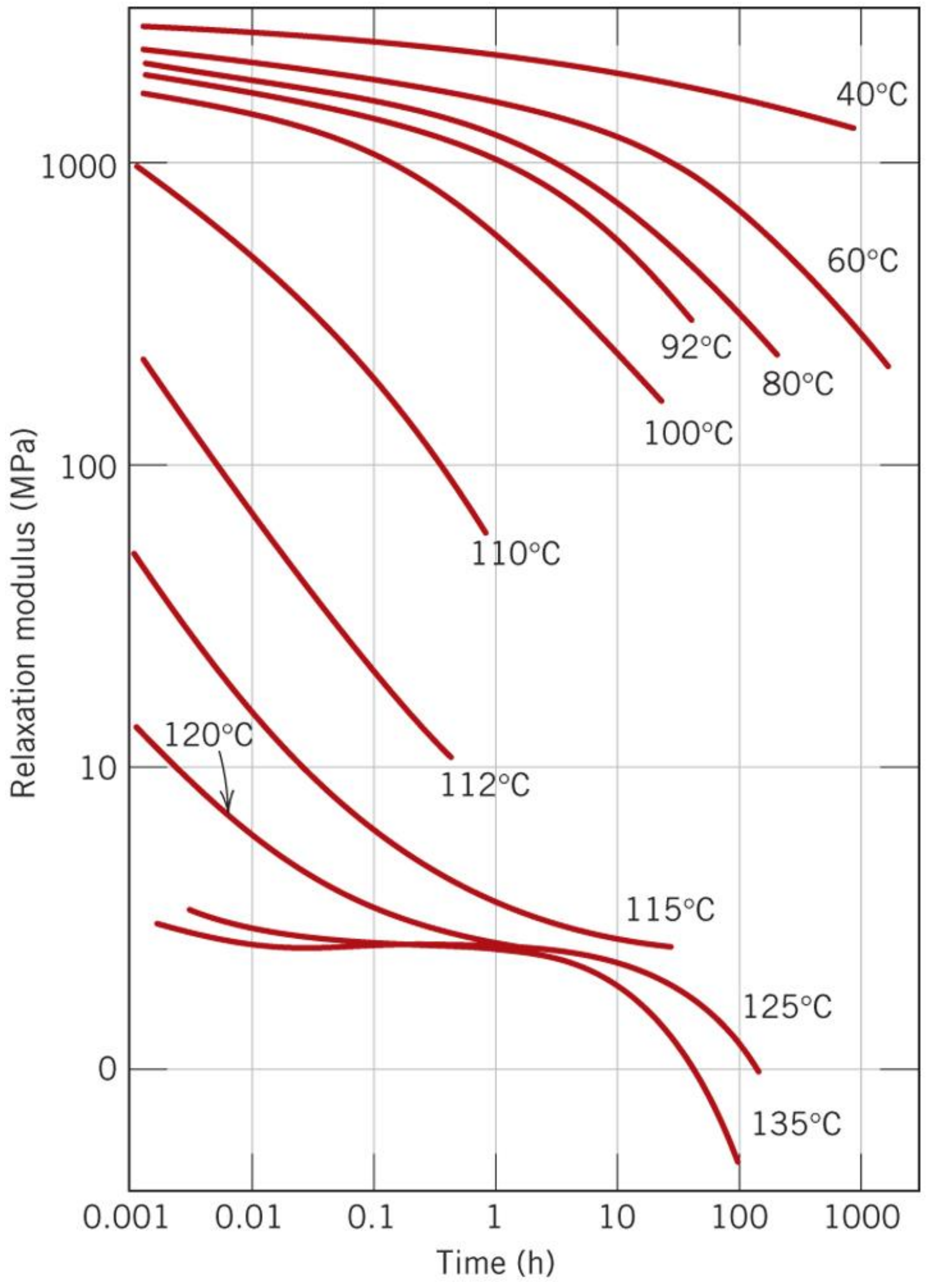
- Q2. When citing the ductility as percent elongation for semicrystalline polymers, it is not necessary to specify the specimen gauge length, as in the case with metals. Why is this so? Write a short answer.

EXAMPLE 2 – ESTIMATING THE TRANSITION GLASS TEMPERATURE TG

- Estimate the T_g for polyisobutylene using the logarithm of $E_r(t)$ vs the logarithm of time for at a variety of temperatures.

Steps:

- 1. Determine the value of relaxation modulus $E_r(10)$ at 10 seconds for each temperature plotted in the graph
- 2. Use the data to make a plot of $E_r(10)$ vs temperature
- 3. Estimate the T_g from the slope of the glass transition region



EXAMPLE 3 – ELASTICITY AND VISCOELASTICITY IN BIOLOGICAL MATERIALS

- Use the self-directed module in bioelasticity created by the University of Cambridge to learn more about viscoelasticity.

This can be accessed using:

https://www.doitpoms.ac.uk/tlplib/bioelasticity/index.php

- Make sure you complete the QUESTIONS section, there is a quiz that will help you improve your knowledge

EXAMPLE 4 – STRAIN TIME PLOTS

- On the basis of the curves in the slide on viscoelastic deformation and the dependency of glass transition temperature on polymer molecular weight (Callister book), sketch schematic strain–time plots for the following polystyrene materials at the specified temperatures:
- (a) Crystalline at 70°C
- (b) Amorphous at 180°C
- (c) Crosslinked at 180°C
- (d) Amorphous at 100°C

EXAMPLE 5 – SEGMENT OF AN OIL PIPE-LINE

- Oil gathering pipe-lines are very small pipelines usually from 2 to 8 inches in diameter in used in areas where crude oil is found deep within the ground. These pipelines are made of steel or plastic components and are designed to carry large amounts of liquid at high temperatures. The plastic parts of a pipeline require to be resistant to corrosion, durable in water, strong acids and organic liquids. Since the pipeline has to be above ground in certain areas, good resistance to solar and UV light is also a requirement.

Using the internet, explore database to select a material that can be used for forming oil pipe parts. The material will preferably made by polymer extrusion as a hollow 3D shape with simple parallel intrusions.

EXAMPLE 6 – GLASS-TRANSITION TEMPERATURE DEMONSTRATIONS

- Examine the effect of temperature on polymers by looking at the different response of a bouncy ball to changes in temperature
- https://www.doitpoms.ac.uk/tlplib/glass-transition/demos.php

- Question:
- Below and above the Tg of a rubber ball it will rebound after impact. At Tg the ball will not rebound very much at all. Explain how the ball bounces in these three cases from an energetic point of view.